Fizika

9-sinf

Mavzu: **Moddaning solishtirma erish issiqligi. Amorf jismlarning erishi va qotishi.**

Moddaning solishtirma erish issiqligi

Erish temperaturasida turgan 1 kg kristall modda to‘liq eritish uchun zarur bo‘lgan issiqlik miqdoriga solishtirma erish issiqligi deyiladi va $\boldsymbol{\lambda}$ ($\boldsymbol{lambda}$) bilan belgilanadi.

$$\lambda = \frac{Q_e}{m} \qquad Q_e = \lambda \cdot m$$

Q_e – erish temperasida moddani suyuqlikka aylantirish uchun zarur bo‘ladigan issiqlik miqdori.

Issiqlik miqdori

Berilgan massali kristall jismni erish temperaturasida suyuqlikka aylantirish uchun qancha issiqlik miqdori sarflangan bo‘lsa, shu temperaturada suyuq holatdan qattiq holatga aylanishida shuncha issiqlik miqdori ajralib chiqadi.

№	**Modda**	λ, kJ / kg	№	**Modda**	λ, kJ / kg
1	Simob	12	6	Kumush	105
2	Qo‘rg‘oshin	25	7	Mis	205
3	Qalay	60	8	Temir	266
4	Oltin	64	9	Muz	334
5	Po‘lat	84	10	Alyuminiy	385

Amorf moddalar

Atom va molekulalari fazoda aniq tartibli vaziyatlarni egallamasa, bunday qattiq moddalar ***amorflar*** deyiladi.

Amorf moddalar



Yantar



Marvarid



Kanifol



Polietilen

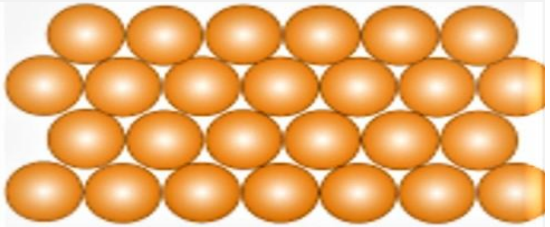


shisha

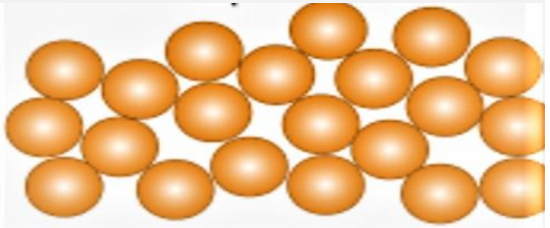


asalari uyasi

Kristall

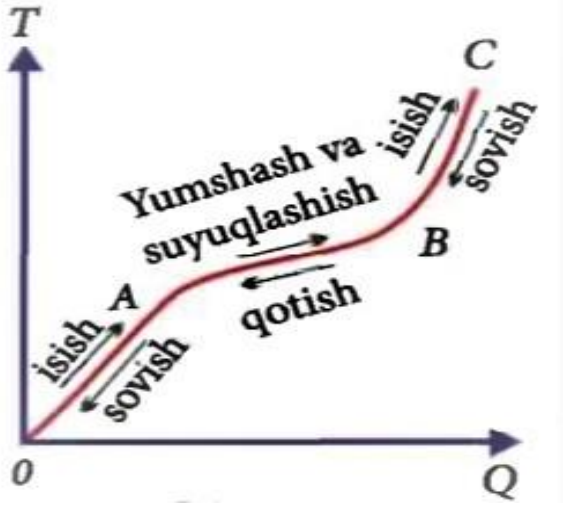


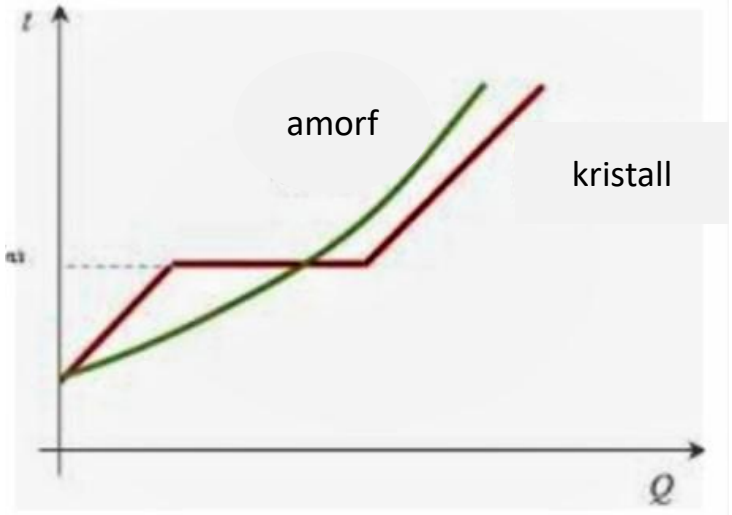
Amorf



Amorf jismlarning erishi va qotishi

Amorf jismlarning erish va qotish grafigi





Amorf jismlar aniq erish temperaturasiga ega emas. Issiqlik berilganda amorf jismlar avval asta-sekin yumshaydi, so‘ngra suyuqlikka o‘ta boshlaydi.

22-mashq 1-masala

Erish temperaturasida turgan 3 kg muzni suvga aylantirish uchun unga qancha issiqlik miqdori berish kerak?

Berilgan:

$m = 3\ kg$

$\lambda = 334\frac{kJ}{kg} = 334 \cdot 10^3 \text{J/kg}$

Topish kerak: $Q - ?$

Formulasi:

$Q = \lambda m$

Yechish: $Q = 334 \cdot \frac{10^3 \text{J}}{\text{kg}} \cdot 3kg = 1002 \cdot 10^3 \text{J} = 1002 \text{kJ}$

Javob: $Q = 1002kJ$

22-mashq 2-masala

Erish temperaturasida turgan m massali qalayni to‘liq eritishga $10\,kJ$ issiqlik miqdori sarflandi. Eritilgan qalayning massasini toping.

Berilgan:

$Q = 10kJ = 10 \cdot 10^3 \text{J}$

$\lambda = 60 \frac{kJ}{kg} = 60 \cdot 10^3 \text{J/kg}$

Topish kerak: m—?

Formulasi:

$Q = \lambda m$

$m = \frac{Q}{\lambda}$

Yechish: $m = \frac{10 \cdot 10^3 J}{60 \cdot 10^3 J/kg} \approx 0{,}17kg$

Javob: $\boldsymbol{m \approx 0,17kg}$

Mustaqil bajarish uchun topshiriqlar

1. Mavzuni o‘qish va savollarga javob yozish.(115-bet)
2. 22-mashqning 3, 4, 5, 6-masalasini yechish.